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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

MINI PROJECT REPORT

BANK CUSTOMER CHURN ANALYSIS

Exploratory Data Analysis, Visualization and Data Preprocessing

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CERTIFICATE

This is to certify that the mini project report titled “Bank Customer Churn Analysis” is submitted by JAISON X (814725243064), JAYAPRAGASH B (814725243066) and NISHANTH M (814725243116), II AI & DS - B, from the Department of Artificial Intelligence and Data Science, SRM TRP Engineering College. The project analyzes bank customer attributes and uses data preprocessing, exploratory data analysis and visualization to examine patterns associated with customer churn.

The report is submitted as part of the academic mini project work under the guidance of V. Ezhil Swanly.

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ABSTRACT

Customer churn, the loss of existing customers who close their accounts or stop using a bank's services, is a major concern for banks because retaining a customer is generally less costly than acquiring a new one. This project, "Bank Customer Churn Analysis," uses a SmartBank customer dataset containing CustomerID, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary and Exited. Python, Pandas, NumPy, Matplotlib and Seaborn are used to inspect and validate the data and to create four visualizations: the overall churn distribution, churn by geography and gender, customer age versus churn status, and a feature correlation heatmap.

The dataset contains 10,000 customer records and 12 columns, with no missing values and no duplicate rows. Of these customers, 8,022 (80.22%) were retained and 1,978 (19.78%) churned, which indicates a moderately imbalanced target. Churn is similar across France, Spain and Germany and across gender groups, and the age distributions of retained and churned customers are nearly identical. The correlation heatmap shows coefficients close to zero (between −0.03 and 0.03) for all numeric feature pairs, including every feature paired with Exited. This is an educational exploratory project; the charts describe sample-level patterns and do not represent a trained or validated churn prediction model.

CHAPTER 1: INTRODUCTION

1.1 Background

Banks hold detailed information about their customers, including demographics, account balances, product usage and engagement. Customer churn occurs when a customer leaves the bank, and a high churn rate reduces revenue and increases the cost of acquiring replacement customers. Exploratory data analysis helps bank analysts understand who their customers are, how many leave, and which attributes may be associated with leaving, before any predictive model is built.

1.2 Problem Statement

Raw customer records must be inspected for data quality issues and summarized before they can support retention decisions. The project addresses this by checking the SmartBank customer dataset for missing values and duplicates, measuring the proportion of customers who churn, and visualizing how churn varies across geography, gender and age, and how the numeric features relate to one another and to churn.

1.3 Objectives

- Inspect the dataset and confirm its shape, data types, missing values and duplicate records.
- Measure the overall churn rate and the class balance between retained and churned customers.
- Compare churn across geography (France, Spain, Germany) and gender.
- Compare the age distribution of retained and churned customers using a box plot.
- Use a correlation heatmap to examine pairwise relationships among numeric features and with churn, and document the limitations of the analysis.

1.4 Scope

The project focuses on data preprocessing and exploratory data analysis of the supplied 10,000-record customer dataset. It does not establish causes of churn, provide financial advice, or present a trained machine-learning model with validated predictive performance.

1.5 Expected Outcome

The expected outcome is a validated customer dataset, a reproducible Google Colab notebook, four visualizations and a summary of sample-level observations that can support further churn analytics and retention planning.

CHAPTER 2: DATASET DESCRIPTION

2.1 Dataset Source

The project uses the SmartBank customer dataset, which is part of the SmartBank – Integrated Banking Management & Customer Churn System project. The notebook loads the data into a Pandas DataFrame for inspection and visualization; the complete notebook is available at the GitHub link given in the appendix.

2.2 Dataset Overview

- Total records: 10,000 customers and 12 columns
- Missing values: 0 in every column
- Duplicate rows: 0
- Target variable: Exited (0 = Retained, 1 = Churned)
- Variables: CustomerID, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary and Exited

Figure 2.1: Dataset Preview

The first five records of the dataset and its shape, (10000, 12), as displayed in the notebook.

Dataset Shape: (10000, 12)

	CustomerID	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	15600000	452	France	Female	61	1	113528.171589	1	0	0	196497.367656	1
1	15600001	785	France	Male	70	6	138786.963707	2	0	0	103499.548919	0
2	15600002	698	France	Female	67	2	233169.998092	1	0	1	111821.652267	0
3	15600003	620	Spain	Female	67	4	132888.460413	1	1	0	126314.621842	0
4	15600004	456	France	Male	21	6	76306.622470	2	1	0	193244.231234	0

2.3 Attribute Description

Attribute	Description	Role
CustomerID	Unique customer identifier	Identifier
CreditScore	Credit score of the customer	Numeric feature
Geography	Country of the customer (France, Spain, Germany)	Categorical feature
Gender	Gender of the customer (Male, Female)	Categorical feature
Age	Customer age in years	Numeric feature
Tenure	Number of years the customer has been with the bank	Numeric feature
Balance	Account balance of the customer	Numeric feature
NumOfProducts	Number of bank products held by the customer	Numeric feature
HasCrCard	Whether the customer has a credit card (1 = Yes, 0 = No)	Binary feature
IsActiveMember	Whether the customer is an active member (1 = Yes, 0 = No)	Binary feature
EstimatedSalary	Estimated salary of the customer	Numeric feature
Exited	Churn status (0 = Retained, 1 = Churned)	Target label

2.4 Dataset Preparation

Geography and Gender are stored as readable text categories, while HasCrCard, IsActiveMember and Exited are stored as 0/1 flags. CustomerID is a unique identifier and carries no analytical meaning, so it is excluded from the correlation analysis. The correlation heatmap uses the numeric and binary columns only; Geography and Gender are analyzed through grouped count plots.

CHAPTER 3: METHODOLOGY

3.1 Project Workflow

1. Load the SmartBank customer dataset into a Pandas DataFrame in Google Colab.
2. Inspect the dataset shape, column names, data types and sample records.
3. Check the dataset for missing values and duplicate rows.
4. Store the validated data as the cleaned DataFrame (clean_df) of 10,000 rows and 12 columns.
5. Plot the overall churn distribution using a count plot.
6. Compare churn across Geography and Gender using grouped count plots.
7. Compare the age distribution of retained and churned customers using a box plot.
8. Generate a correlation heatmap of the numeric features and the churn label.
9. Interpret the charts cautiously and record project limitations and future work.

3.2 Tools and Technologies

Python in Google Colab was used as the development environment. Pandas and NumPy were used for tabular data processing and numerical operations, Matplotlib and Seaborn for visualization, and GitHub for hosting the project notebook.

CHAPTER 4: DATA PREPROCESSING

4.1 Data Inspection

The notebook reads the dataset and checks its shape and first records. The dataset contains 10,000 rows and 12 columns, covering demographic details (Geography, Gender, Age), account details (CreditScore, Tenure, Balance, NumOfProducts, HasCrCard), engagement (IsActiveMember), income (EstimatedSalary) and the churn label (Exited).

4.2 Missing Values and Duplicates

A column-wise count of missing values and a count of duplicate rows were computed on the raw data. Every column reports zero missing values and the dataset contains zero duplicate rows, as shown in Figure 4.1.

Figure 4.1: Missing Values and Duplicate Check

The output confirms that no values are missing, no rows are duplicated, and the cleaned shape remains (10000, 12).

```
Missing values in raw data:
 CustomerID         0
CreditScore         0
Geography           0
Gender              0
Age                 0
Tenure              0
Balance             0
NumOfProducts       0
HasCrCard           0
IsActiveMember      0
EstimatedSalary     0
Exited              0
dtype: int64
Duplicate rows: 0

Cleaned Data Shape: (10000, 12)
```

4.3 Cleaning and Preparation

Because no missing values or duplicate records were found, no rows were removed and no imputation was required. The cleaned dataset (clean_df) therefore retains all 10,000 records and 12 columns.

4.4 Categorical and Binary Variables

Geography and Gender are categorical text columns and are plotted directly as count plots grouped by churn status. HasCrCard, IsActiveMember and Exited are binary flags; they are included in the correlation matrix as 0/1 values, which should be read as measures of linear association with a binary variable rather than as continuous measurements.

4.5 Feature Review

The features reviewed are CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary and Exited. The notebook uses these columns to create group comparisons, a box plot and a correlation matrix. CustomerID is retained in the data but is not used as an analytical feature.

4.6 Clean Dataset Summary

The cleaned dataset contains 10,000 records, 12 fields and no missing or duplicated records. Its size is suitable for stable exploratory summaries and provides a sound basis for any later modeling work.

CHAPTER 5: EXPLORATORY DATA ANALYSIS

5.1 Dataset Summary

Metric	Result
Total customer records	10,000
Number of columns	12
Missing values	0 across all columns
Duplicate rows	0
Retained customers (Exited = 0)	8,022 (80.22%)
Churned customers (Exited = 1)	1,978 (19.78%)
Retained-to-churned ratio	approximately 4.06 : 1

5.2 Churn Distribution

The count plot shows 8,022 retained customers and 1,978 churned customers. Roughly one in five customers (19.78%) has left the bank, so the target is imbalanced: any later classification model must be evaluated with metrics such as precision, recall and F1-score rather than accuracy alone.

5.3 Churn by Geography and Gender

The grouped count plots compare retained and churned customers in each country and gender group. France has about half of all customers, while Spain and Germany each have roughly a quarter. In every country and for both genders, churned customers form roughly one-fifth of the group, so no single segment stands out in this chart.

5.4 Customer Age versus Churn

The box plot compares the age distribution of retained and churned customers. Both groups have a median age of about 46 years, similar interquartile ranges (about 32 to 60 years) and the same overall range of about 18 to 74 years, indicating that age alone does not visibly separate the two groups.

5.5 Visualizations

Figure 5.1: Customer Churn Distribution

The count plot compares the number of retained customers (0) with the number of churned customers (1).

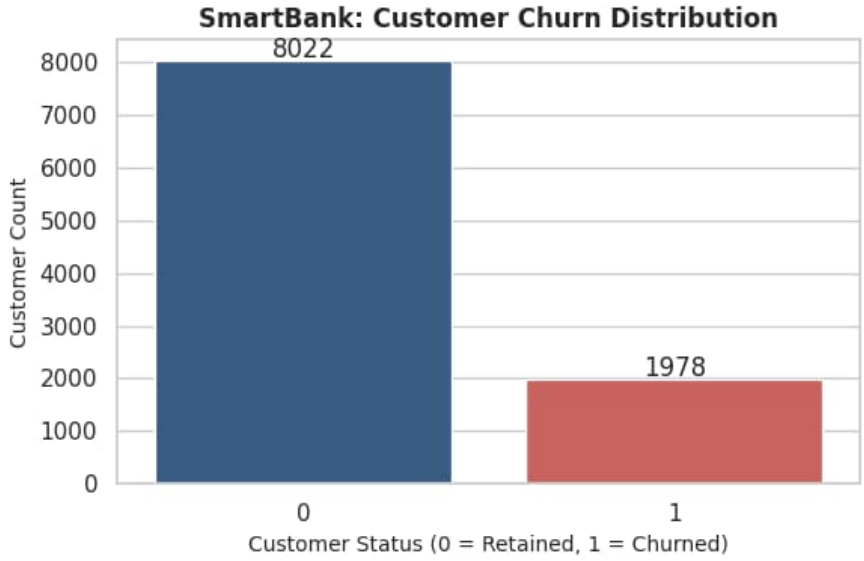


Figure 5.2: Customer Churn by Geography and Gender

The grouped count plots compare retained and churned customers across the three countries and the two gender groups.

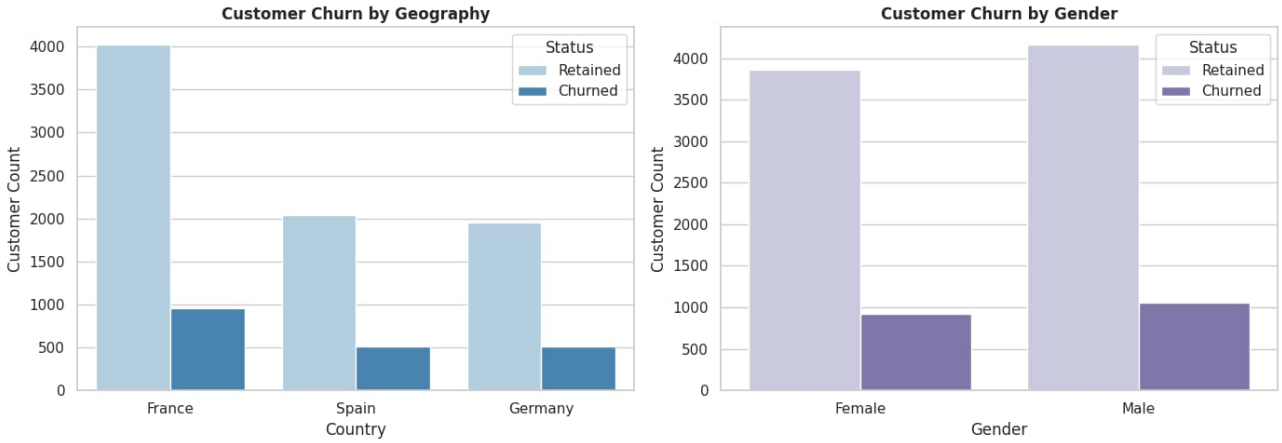
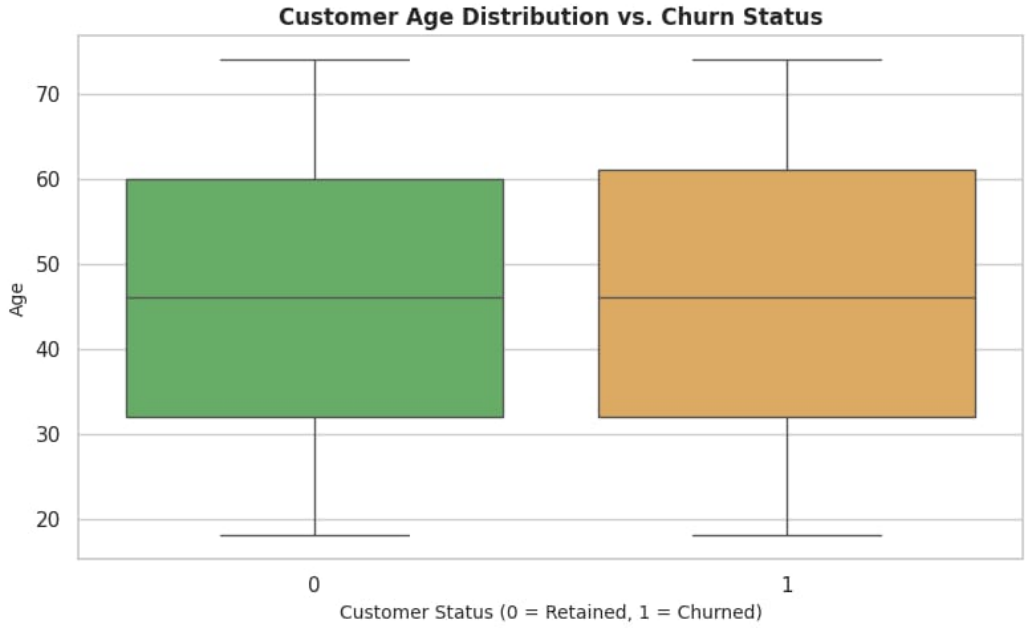


Figure 5.3: Customer Age Distribution versus Churn Status

The box plot compares the age distribution of retained (0) and churned (1) customers.



CHAPTER 6: RESULTS AND INTERPRETATION

6.1 Churn Distribution

- Out of 10,000 customers, 8,022 were retained and 1,978 churned, giving a churn rate of 19.78%. The retained class is roughly four times larger than the churned class.

6.2 Geography and Gender

- The churn share is close to 20% in France, Spain and Germany, so geography shows no strong difference in this sample.
- Churn is similar for female and male customers (close to 20% in both groups), although the dataset contains slightly more male customers.

6.3 Age and Churn

- The median age of retained and churned customers is nearly identical (about 46 years), and both groups span about 18 to 74 years. Age alone does not distinguish churned customers from retained customers.

6.4 Correlation Heatmap

- The heatmap displays pairwise Pearson correlation coefficients among CreditScore, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary and Exited.
- All off-diagonal coefficients lie between −0.03 and 0.03. The largest in magnitude are Age with EstimatedSalary (−0.03), CreditScore with Tenure (0.02) and Balance with IsActiveMember (0.02).

- Exited has coefficients of 0.00 to 0.01 (in magnitude) with every feature, so no single numeric feature is linearly related to churn in this dataset.

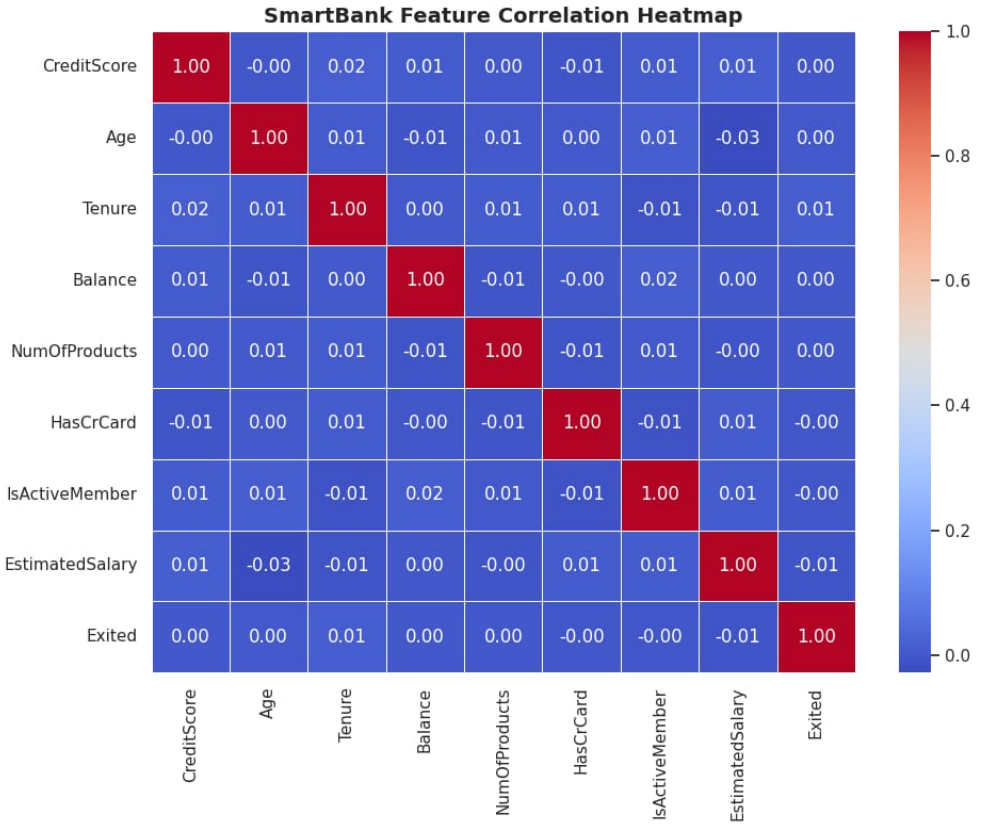
6.5 Interpretation

- The visualizations make it easier to compare distributions and relationships in the supplied records. Correlation measures linear association, not causation, and a coefficient near zero does not rule out non-linear or interaction effects. The results should therefore be treated as exploratory observations only.

6.6 Project Output

Figure 6.1: SmartBank Feature Correlation Heatmap

The heatmap shows the correlation among the numeric features and the churn label; coefficients are close to zero throughout.



The project output consists of the validated customer dataset, the Google Colab notebook and four charts: churn distribution, churn by geography and gender, age versus churn, and the correlation heatmap. These outputs demonstrate an end-to-end introductory workflow for bank customer data inspection, preprocessing and visualization.

CHAPTER 7: KEY FINDINGS AND RECOMMENDATIONS

7.1 Key Findings

- The dataset contains 10,000 customer records and 12 variables, with no missing values and no duplicates.
- 19.78% of customers (1,978) churned and 80.22% (8,022) were retained, so the target is moderately imbalanced.
- Churn is close to one-fifth in every country and for both genders in the displayed charts.
- Retained and churned customers have nearly identical age distributions (median about 46 years).
- All pairwise correlations are between −0.03 and 0.03, including every correlation with churn.

7.2 Recommendations

- Because linear correlations with churn are negligible, evaluate non-linear models (for example decision trees, random forests or gradient boosting) and feature interactions before drawing conclusions.
- Handle the class imbalance with stratified splitting, class weights or resampling, and evaluate models with precision, recall, F1-score, ROC-AUC and a confusion matrix.
- Maintain a data dictionary describing every field, its unit and its allowed values.
- Enrich the data with behavioral information such as transaction frequency, complaints and service usage, which is more likely to explain churn than static profile fields.
- Validate the findings on real, representative customer data and obtain business review before using any result for customer-retention decisions.

CHAPTER 8: LIMITATIONS AND FUTURE ENHANCEMENTS

8.1 Limitations

- The analysis is exploratory and descriptive; it does not include a trained or evaluated machine-learning model.
- Pearson correlation captures linear relationships only, and Geography and Gender are not part of the heatmap.
- The dataset is a single snapshot with no transaction history or time dimension, so churn drivers over time cannot be studied.
- The near-zero correlations and evenly spread numeric features suggest that the dataset behaves like a simulated or benchmark dataset, so patterns may not generalize to a real bank.
- Group differences in the charts are read visually; no statistical significance tests were performed.

8.2 Future Enhancements

- Train and compare classification models such as logistic regression, random forest and gradient boosting using a proper train/test split and cross-validation.
- Apply feature engineering (for example balance-to-salary ratio, age groups and product-usage indicators) and encode categorical variables for modeling.

- Use statistical tests such as chi-square and t-tests to check whether group differences are significant.
- Explain model predictions with feature-importance methods and build a retention dashboard for bank managers.

CHAPTER 9: CONCLUSION

This mini project demonstrates a basic bank customer churn analysis workflow using Python, Pandas, NumPy, Matplotlib and Seaborn. The SmartBank dataset of 10,000 customers was inspected and found to be free of missing values and duplicates, and four visualizations were used to explore the churn distribution, geography, gender, age and feature correlations. About 19.78% of customers churned, churn was similar across countries, gender groups and ages, and no numeric feature showed a meaningful linear correlation with churn. These findings are exploratory only. A dependable churn prediction system would require non-linear modeling, richer behavioral data, proper handling of class imbalance, and validation on representative customer data.

REFERENCES

- SmartBank customer dataset (10,000 records, 12 columns) used in the EDA assignment notebook.
- Pandas documentation: https://pandas.pydata.org/docs/
- NumPy documentation: https://numpy.org/doc/stable/
- Matplotlib documentation: https://matplotlib.org/stable/
- Seaborn documentation: https://seaborn.pydata.org/

APPENDIX: PROJECT FILES AND VISUALIZATIONS

The project materials used for this report include the Google Colab notebook, the customer dataset, and the visualization outputs.

- EDA assignment.ipynb — Google Colab notebook
- churn_distribution.png — Customer churn distribution
- churn_by_geography.png — Customer churn by geography and gender
- age_vs_churn.png — Customer age distribution versus churn status
- correlation_heatmap.png — Feature correlation heatmap

GITHUB LINK:

https://github.com/814725243064-jxc/SmartBank---Integrated-Banking-Management-Customer-Churn-System/blob/main/EDA%20assignment.ipynb